

Advanced Engineering Mathematics

Vector Calculus

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1 Vector Functions

1.1 Definition and Representation of Vector Functions

Key Concept

A vector-valued function assigns a vector to each scalar input:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

A vector function describes motion or geometry in space. Each component function determines behavior along one axis.

Example

Write the vector function for motion where $x = t$, $y = t^2$, $z = t^3$.

Solution:

Given motion where $x = t$, $y = t^2$, $z = t^3$:

$$\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$$

Example

Evaluate $\mathbf{r}(2)$ for $\mathbf{r}(t) = \langle t^2, \sin t, e^t \rangle$.

Solution:

$$\begin{aligned}\mathbf{r}(2) &= \langle 2^2, \sin 2, e^2 \rangle \\ &= \langle 4, \sin 2, e^2 \rangle\end{aligned}$$

Example

Determine if $\mathbf{r}(t) = \langle \cos t, \sin t, 1 \rangle$ lies on a cylinder.

Solution:

$$\begin{aligned}x^2 + y^2 &= \cos^2 t + \sin^2 t \\ &= 1\end{aligned}$$

Thus, the curve lies on the cylinder $x^2 + y^2 = 1$.

1.2 Limits and Continuity of Vector Functions

Key Concept

A vector function is continuous if each component function is continuous:

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \langle \lim x(t), \lim y(t), \lim z(t) \rangle$$

Example

Evaluate $\lim_{t \rightarrow 0} \langle \sin t/t, t^2, e^t \rangle$.

Solution:

$$\lim_{t \rightarrow 0} \langle \frac{\sin t}{t}, t^2, e^t \rangle = \langle 1, 0, 1 \rangle$$

Example

Determine continuity of $\mathbf{r}(t) = \langle t^2, \sqrt{t}, \ln t \rangle$.

Solution:

$$\begin{aligned} t^2 &\text{ is continuous for all } t \\ \sqrt{t} &\text{ is continuous for } t \geq 0 \\ \ln t &\text{ is continuous for } t > 0 \end{aligned}$$

Thus, the function is continuous only for $t > 0$.

1.3 Differentiation of Vector Functions

Key Concept

The derivative is taken component-wise:

$$\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

Example

Differentiate $\mathbf{r}(t) = \langle t^2, \sin t, e^t \rangle$.

Solution:

$$\mathbf{r}'(t) = \langle 2t, \cos t, e^t \rangle$$

Example

Find velocity and acceleration for $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$.

Solution:

$$\mathbf{v}(t) = \langle 1, 2t, 3t^2 \rangle$$

$$\mathbf{a}(t) = \langle 0, 2, 6t \rangle$$

Example

Find tangent vector at $t = 1$ for $\mathbf{r}(t) = \langle \ln t, t^2, e^t \rangle$.

Solution:

$$\mathbf{r}'(t) = \left\langle \frac{1}{t}, 2t, e^t \right\rangle$$

$$\mathbf{r}'(1) = \langle 1, 2, e \rangle$$

1.4 Integration of Vector Functions

Key Concept

Integration is performed component-wise:

$$\int \mathbf{r}(t) dt = \left\langle \int x(t) dt, \int y(t) dt, \int z(t) dt \right\rangle$$

Example

Integrate $\mathbf{r}(t) = \langle t, 2t, 3t \rangle$.

Solution:

$$\int \mathbf{r}(t) dt = \left\langle \frac{t^2}{2}, t^2, \frac{3t^2}{2} \right\rangle + \mathbf{C}$$

Example

Solve for $\mathbf{r}(t)$ given $\mathbf{r}'(t) = \langle 2t, \cos t, e^t \rangle$.

Solution:

$$\begin{aligned}\mathbf{r}(t) &= \int \langle 2t, \cos t, e^t \rangle dt \\ &= \langle t^2, \sin t, e^t \rangle + \mathbf{C}\end{aligned}$$

1.5 Arc Length and Speed

Key Concept

Speed is magnitude of velocity:

$$|\mathbf{r}'(t)|$$

Arc length:

$$L = \int_a^b |\mathbf{r}'(t)| dt$$

Example

Find speed for $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$.

Solution:

$$\begin{aligned}\mathbf{r}'(t) &= \langle -\sin t, \cos t, 1 \rangle \\ |\mathbf{r}'(t)| &= \sqrt{\sin^2 t + \cos^2 t + 1} \\ &= \sqrt{2}\end{aligned}$$

Example

Find arc length from 0 to 1.

Solution:

$$\begin{aligned}L &= \int_0^1 \sqrt{2} dt \\ &= \sqrt{2}\end{aligned}$$

1.6 Unit Tangent, Normal, and Curvature

Key Concept

Unit tangent:

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Example

Find unit tangent for $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$.

Solution:

$$\begin{aligned}\mathbf{r}'(t) &= \langle 1, 2t, 0 \rangle \\ |\mathbf{r}'(t)| &= \sqrt{1 + 4t^2} \\ \mathbf{T}(t) &= \frac{\langle 1, 2t, 0 \rangle}{\sqrt{1 + 4t^2}}\end{aligned}$$

1.7 Vector Functions in Engineering Applications

Vector functions model trajectories, fluid motion, and signal propagation paths.

Example

The vector function

$$\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$$

represents a helix. The projection onto the xy -plane is circular motion, while $z = t$ introduces linear growth, forming a spiral trajectory common in electromagnetic wave propagation.

Example

In signal processing, parametric curves such as:

$$\mathbf{r}(t) = \langle x(t), y(t) \rangle$$

can represent phase-space trajectories of signals, where $x(t)$ is the signal and $y(t)$ its derivative, enabling analysis of system dynamics.

2 Partial Derivatives

2.1 Definition of Partial Derivatives

Key Concept

A partial derivative measures the rate of change of a multivariable function with respect to one variable while holding all other variables constant:

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y, z) - f(x, y, z)}{h}$$

Partial derivatives extend the idea of ordinary derivatives to higher dimensions. Instead of a slope along a line, we now measure slopes along coordinate directions.

Example

Find $\frac{\partial f}{\partial x}$ for $f(x, y) = x^2y + 3y^2$.

Solution:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(x^2y + 3y^2) \\ &= 2xy + 0 \\ &= 2xy\end{aligned}$$

Example

Find $\frac{\partial f}{\partial y}$ for the same function.

Solution:

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(x^2y + 3y^2) \\ &= x^2 + 6y\end{aligned}$$

Example

Evaluate $\frac{\partial f}{\partial x}$ at $(1, 2)$.

Solution:

$$\frac{\partial f}{\partial x} = 2xy = 2(1)(2) = 4$$

2.2 Higher-Order Partial Derivatives

Key Concept

Higher-order partial derivatives are obtained by differentiating multiple times:

$$\frac{\partial^2 f}{\partial x^2}, \quad \frac{\partial^2 f}{\partial x \partial y}$$

Example

Find $\frac{\partial^2 f}{\partial x^2}$ for $f = x^2y$.

$$\begin{aligned}\frac{\partial f}{\partial x} &= 2xy \\ \frac{\partial^2 f}{\partial x^2} &= 2y\end{aligned}$$

Example

$$f(x, y) = x^2y$$

We are asked to compute the mixed partial derivative:

$$\frac{\partial^2 f}{\partial y \partial x}$$

Step 1: Differentiate with respect to x

Treat y as a constant:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(x^2y) \\ &= y \cdot \frac{d}{dx}(x^2) \\ &= y(2x) \\ &= 2xy\end{aligned}$$

Step 2: Differentiate the result with respect to y

Now treat x as a constant:

$$\begin{aligned}\frac{\partial}{\partial y}(2xy) &= 2x \cdot \frac{d}{dy}(y) \\ &= 2x\end{aligned}$$

Final Answer:

$$\frac{\partial^2 f}{\partial y \partial x} = 2x$$

Example

Let:

$$f(x, y) = xy^2$$

We will compute both:

$$\frac{\partial^2 f}{\partial x \partial y} \quad \text{and} \quad \frac{\partial^2 f}{\partial y \partial x}$$

Step 1: Compute $\frac{\partial^2 f}{\partial x \partial y}$

First differentiate with respect to y :

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(xy^2) \\ &= x \cdot \frac{d}{dy}(y^2) \\ &= x(2y) \\ &= 2xy\end{aligned}$$

Now differentiate with respect to x :

$$\frac{\partial}{\partial x}(2xy) = 2y$$

Thus:

$$\frac{\partial^2 f}{\partial x \partial y} = 2y$$

—

Step 2: Compute $\frac{\partial^2 f}{\partial y \partial x}$

First differentiate with respect to x :

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(xy^2) \\ &= y^2\end{aligned}$$

Now differentiate with respect to y :

$$\frac{\partial}{\partial y}(y^2) = 2y$$

Thus:

$$\frac{\partial^2 f}{\partial y \partial x} = 2y$$

—
Final Conclusion:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 2y$$

Important Insight (Clairaut's Theorem): If a function has continuous second-order partial derivatives, then the mixed partial derivatives are equal:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

2.3 Geometric Interpretation

Key Concept

Partial derivatives represent slopes of tangent lines to curves formed by slicing the surface:

$$z = f(x, y)$$

Example

For $z = x^2 + y^2$, compute slopes:

$$\frac{\partial z}{\partial x} = 2x, \quad \frac{\partial z}{\partial y} = 2y$$

At (1, 1):

$$\frac{\partial z}{\partial x} = 2, \quad \frac{\partial z}{\partial y} = 2$$

Interpretation: slope is steeper as distance from origin increases.

2.4 The Gradient Vector

Key Concept

The gradient is a vector of partial derivatives:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

Example

Let the function be:

$$f(x, y, z) = x^2 + y^2 + z^2$$

—

Step 1: Recall the definition of the gradient

The gradient of a scalar function is:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

So we compute each partial derivative one at a time.

—

Step 2: Compute partial derivatives

Partial with respect to x :

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(x^2 + y^2 + z^2) \\ &= 2x + 0 + 0 \\ &= 2x \end{aligned}$$

Partial with respect to y :

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(x^2 + y^2 + z^2) \\ &= 0 + 2y + 0 \\ &= 2y\end{aligned}$$

Partial with respect to z :

$$\begin{aligned}\frac{\partial f}{\partial z} &= \frac{\partial}{\partial z}(x^2 + y^2 + z^2) \\ &= 0 + 0 + 2z \\ &= 2z\end{aligned}$$

—

Step 3: Form the gradient vector

$$\nabla f = \langle 2x, 2y, 2z \rangle$$

—

Step 4: Evaluate at the point $(1, 2, 3)$

Substitute $x = 1$, $y = 2$, $z = 3$:

$$\begin{aligned}\nabla f(1, 2, 3) &= \langle 2(1), 2(2), 2(3) \rangle \\ &= \langle 2, 4, 6 \rangle\end{aligned}$$

—

Step 5: Compute the magnitude of the gradient

The magnitude of a vector $\langle a, b, c \rangle$ is:

$$|\mathbf{v}| = \sqrt{a^2 + b^2 + c^2}$$

Apply this to $\nabla f(1, 2, 3) = \langle 2, 4, 6 \rangle$:

$$\begin{aligned}
 |\nabla f| &= \sqrt{2^2 + 4^2 + 6^2} \\
 &= \sqrt{4 + 16 + 36} \\
 &= \sqrt{56}
 \end{aligned}$$

Final Answers:

$$\begin{aligned}
 \nabla f &= \langle 2x, 2y, 2z \rangle \\
 \nabla f(1, 2, 3) &= \langle 2, 4, 6 \rangle \\
 |\nabla f(1, 2, 3)| &= \sqrt{56}
 \end{aligned}$$

Geometric Interpretation:

- The gradient vector points in the direction of **maximum increase** of the function.
- At the point $(1, 2, 3)$, the function increases fastest in the direction $\langle 2, 4, 6 \rangle$.
- The magnitude $\sqrt{56}$ tells us **how fast** the function increases in that direction.

2.5 Directional Dependence

Key Concept

Partial derivatives form the foundation for directional derivatives and field analysis. In engineering:

- Gradients represent signal intensity changes
- Partial derivatives describe system sensitivity

Example

For a scalar field such as temperature:

$$T(x, y, z) = x^2 + y^2 + z^2$$

- ∇T gives the direction of **steepest temperature increase**

- $|\nabla T|$ gives the **rate of increase**

This concept directly appears in:

- heat flow analysis
- electromagnetic field intensity
- signal gradient detection in image processing

3 Directional Derivatives

3.1 Definition of Directional Derivatives

Key Concept

The directional derivative of a scalar function $f(x, y, z)$ in the direction of a unit vector $\mathbf{u}$ is defined as:

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$$

The directional derivative measures how a function changes as we move in a specific direction. Unlike partial derivatives, which only measure change along coordinate axes, directional derivatives allow us to measure change in any direction.

Example

Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 1, 0 \rangle$.

Step 1: Normalize the direction vector

$$|\mathbf{v}| = \sqrt{1^2 + 0^2} = 1$$

Thus, $\mathbf{u} = \langle 1, 0 \rangle$

Step 2: Compute the gradient

$$\nabla f = \langle 2x, 2y \rangle$$

Step 3: Evaluate at $(1, 1)$

$$\nabla f(1, 1) = \langle 2, 2 \rangle$$

Step 4: Compute dot product

$$D_{\mathbf{u}}f = \langle 2, 2 \rangle \cdot \langle 1, 0 \rangle = 2$$

3.2 Maximum and Minimum Directional Derivatives

Key Concept

The directional derivative satisfies:

$$\max D_{\mathbf{u}}f = |\nabla f| \quad (\text{in direction of } \nabla f)$$

$$\min D_{\mathbf{u}}f = -|\nabla f| \quad (\text{in direction opposite to } \nabla f)$$

Example

Find the maximum rate of change of $f = x^2 + y^2$ at $(1, 1)$. **Step 1: Compute gradient**

$$\nabla f = \langle 2x, 2y \rangle$$

Step 2: Evaluate at $(1, 1)$

$$\nabla f = \langle 2, 2 \rangle$$

Step 3: Compute magnitude

$$|\nabla f| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

Final Answer: Maximum rate of change is $2\sqrt{2}$.

Example

Find the direction of maximum increase. **Solution:**

$$\mathbf{u} = \frac{\nabla f}{|\nabla f|} = \frac{\langle 2, 2 \rangle}{2\sqrt{2}} = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$$

Example

Find the minimum rate of change of $f = x^2 + y^2$ at $(1, 1)$. **Step 1: Compute the gradient**

$$\nabla f = \langle 2x, 2y \rangle$$

Step 2: Evaluate at $(1, 1)$

$$\nabla f = \langle 2, 2 \rangle$$

Step 3: Compute the magnitude

$$|\nabla f| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

Step 4: Determine the minimum value

$$\min D_{\mathbf{u}}f = -|\nabla f| = -2\sqrt{2}$$

Final Answer: Minimum rate of change is $-2\sqrt{2}$.

Example

Find the direction of minimum change.

Step 1: Recall direction of maximum increase

$$\mathbf{u}_{\max} = \frac{\nabla f}{|\nabla f|} = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$$

Step 2: Reverse the direction

$$\mathbf{u}_{\min} = -\mathbf{u}_{\max} = \left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle$$

Final Answer: Direction of steepest decrease is

$$\left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle$$

Key Insight:

- The gradient ∇f always points in the direction of **maximum increase**.

- The opposite direction $-\nabla f$ points toward **maximum decrease**.
- The magnitude $|\nabla f|$ gives how fast the function increases or decreases.

Geometric Interpretation:

At any point on a surface:

- Moving along $\nabla f \rightarrow$ climb uphill fastest
- Moving opposite $\nabla f \rightarrow$ go downhill fastest

One-line Summary:

Maximum change $= \nabla f $, Minimum change $= - \nabla f $

3.3 Directional Derivative in Three Dimensions

Key Concept

For $f(x, y, z)$, the directional derivative is:

$$D_{\mathbf{u}}f = f_x u_1 + f_y u_2 + f_z u_3$$

Example

Find the directional derivative of $f = x^2 + y^2 + z^2$ at $(1, 2, 2)$ in the direction of $\mathbf{v} = \langle 1, 2, 2 \rangle$.

Step 1: Normalize direction vector

$$|\mathbf{v}| = \sqrt{1 + 4 + 4} = 3$$

$$\mathbf{u} = \left\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right\rangle$$

Step 2: Compute gradient

$$\nabla f = \langle 2x, 2y, 2z \rangle$$

Step 3: Evaluate at $(1, 2, 2)$

$$\nabla f = \langle 2, 4, 4 \rangle$$

Step 4: Dot product

$$\begin{aligned} D_{\mathbf{u}}f &= \langle 2, 4, 4 \rangle \cdot \left\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right\rangle \\ &= \frac{2}{3} + \frac{8}{3} + \frac{8}{3} \\ &= \frac{18}{3} = 6 \end{aligned}$$

3.4 Geometric Interpretation

Key Concept

The directional derivative represents the slope of the surface in a given direction.

Example

For $f = x^2 + y^2$, moving outward from origin increases value fastest.
Directional derivative is highest along radial direction.

Directional derivatives appear in:

- heat transfer (temperature gradients)
- fluid flow (velocity change along direction)
- signal processing (gradient-based edge detection)

Example

Temperature field:

$$T(x, y) = x^2 + y^2$$

Directional derivative gives how fast temperature changes in a chosen direction.

4 Curl and Divergence

4.1 Vector Fields

Key Concept

A vector field assigns a vector to every point in space:

$$\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$$

Vector fields are used to model physical quantities such as velocity of a fluid, electric fields, and force fields.

Example

Determine the value of the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$ at $(1, 2, 3)$.

$$\mathbf{F}(1, 2, 3) = \langle 1, 2, 3 \rangle$$

Interpretation: At the point $(1, 2, 3)$, the vector points outward from the origin.

4.2 Divergence of a Vector Field

Key Concept

The divergence measures how much a vector field spreads out (source) or converges (sink):

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Example

Find the divergence of $\mathbf{F} = \langle x, y, z \rangle$.

Step 1: Identify components

$$P = x, \quad Q = y, \quad R = z$$

Step 2: Compute partial derivatives

$$\frac{\partial P}{\partial x} = 1, \quad \frac{\partial Q}{\partial y} = 1, \quad \frac{\partial R}{\partial z} = 1$$

Step 3: Sum them

$$\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3$$

Interpretation: The field has a constant positive divergence, meaning it behaves like a uniform source everywhere.

Example

Find divergence of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$.

$$\begin{aligned}\nabla \cdot \mathbf{F} &= \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z^2) \\ &= 2x + 2y + 2z\end{aligned}$$

Interpretation: Divergence varies with position; stronger away from origin.

4.3 Curl of a Vector Field

Key Concept

The curl measures the tendency of a vector field to rotate:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Example

Find the curl of $\mathbf{F} = \langle y, -x, 0 \rangle$.

Step 1: Identify components

$$P = y, \quad Q = -x, \quad R = 0$$

Step 2: Compute determinant

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & -x & 0 \end{vmatrix}$$

Step 3: Expand determinant

$$\begin{aligned}\nabla \times \mathbf{F} &= \mathbf{i}(0 - 0) - \mathbf{j}(0 - 0) + \mathbf{k}(-1 - 1) \\ &= \langle 0, 0, -2 \rangle\end{aligned}$$

Interpretation: The field rotates about the z -axis.

Example

We are given the vector field:

$$\mathbf{F}(x, y, z) = \langle P, Q, R \rangle = \langle xz, yz, xy \rangle$$

Find curl of $\mathbf{F} = \langle xz, yz, xy \rangle$.

Step 1: Recall the definition of curl

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Expanding this determinant gives:

$$\nabla \times \mathbf{F} = \left\langle \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle$$

Step 2: Identify the components

$$P = xz, \quad Q = yz, \quad R = xy$$

Step 3: Compute each component carefully

First component:

$$\begin{aligned}\frac{\partial R}{\partial y} &= \frac{\partial}{\partial y}(xy) = x \\ \frac{\partial Q}{\partial z} &= \frac{\partial}{\partial z}(yz) = y \\ \Rightarrow \text{First component} &= x - y\end{aligned}$$

Second component:

$$\frac{\partial P}{\partial z} = \frac{\partial}{\partial z}(xz) = x$$
$$\frac{\partial R}{\partial x} = \frac{\partial}{\partial x}(xy) = y$$

$$\Rightarrow \text{Second component} = x - y$$

Third component:

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(yz) = 0$$
$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(xz) = 0$$

$$\Rightarrow \text{Third component} = 0 - 0 = 0$$

Step 4: Combine results

$$\nabla \times \mathbf{F} = \langle x - y, x - y, 0 \rangle$$

Final Answer:

$$\nabla \times \mathbf{F} = \langle x - y, x - y, 0 \rangle$$

Key Concept

Divergence and curl provide physical meaning:

- Divergence: source or sink strength
- Curl: rotational tendency

Example

Fluid flow interpretation:

- If $\nabla \cdot \mathbf{v} > 0$: fluid expands (source)
- If $\nabla \cdot \mathbf{v} < 0$: fluid compresses (sink)

Rotation example:

A whirlpool has nonzero curl, indicating rotation in the flow field.

Key Concept

A vector field is:

- Conservative if $\nabla \times \mathbf{F} = 0$
- Solenoidal if $\nabla \cdot \mathbf{F} = 0$

Example

Check if $\mathbf{F} = \langle 2x, 2y, 2z \rangle$ is conservative.

$$\nabla \times \mathbf{F} = \mathbf{0}$$

Thus, it is conservative.

Example

Check if $\mathbf{F} = \langle -y, x, 0 \rangle$ is solenoidal.

$$\nabla \cdot \mathbf{F} = 0$$

Thus, it has no sources or sinks.

5 Integral Concepts in Vector Calculus

5.1 Line Integrals

Key Concept

A line integral of a vector field along a curve C is defined as:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

This represents the accumulation of a quantity along a path. In physics, it commonly represents **work done by a force**.

To evaluate a line integral, we parametrize the curve:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$

Example

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle x, y \rangle$ and C is the line from $(0, 0)$ to $(1, 1)$.

Step 1: Parametrize the curve

$$\mathbf{r}(t) = \langle t, t \rangle, \quad 0 \leq t \leq 1$$

Step 2: Compute derivative

$$\mathbf{r}'(t) = \langle 1, 1 \rangle$$

Step 3: Evaluate vector field along path

$$\mathbf{F}(\mathbf{r}(t)) = \langle t, t \rangle$$

Step 4: Compute dot product

$$\mathbf{F} \cdot \mathbf{r}' = t(1) + t(1) = 2t$$

Step 5: Integrate

$$\int_0^1 2t \, dt = [t^2]_0^1 = 1$$

Final Answer: 1

Example

Evaluate the work done by the vector field

$$\mathbf{F}(x, y, z) = \langle yz, xz, xy \rangle$$

along the curve

$$\mathbf{r}(t) = \langle t, t^2, t^3 \rangle, \quad 0 \leq t \leq 1.$$

Step 1: Recall the formula for work

The work done by a vector field along a curve is given by the line integral

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Thus, we must first compute $\mathbf{r}'(t)$ and then evaluate the vector field along the path.

Step 2: Differentiate the parametrization

Since

$$\mathbf{r}(t) = \langle t, t^2, t^3 \rangle,$$

we differentiate component-wise:

$$\begin{aligned} \mathbf{r}'(t) &= \left\langle \frac{d}{dt}(t), \frac{d}{dt}(t^2), \frac{d}{dt}(t^3) \right\rangle \\ &= \langle 1, 2t, 3t^2 \rangle \end{aligned}$$

Step 3: Evaluate the vector field along the curve

The field is

$$\mathbf{F}(x, y, z) = \langle yz, xz, xy \rangle$$

Along the path $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$, we substitute

$$x = t, \quad y = t^2, \quad z = t^3$$

Therefore,

$$\begin{aligned} \mathbf{F}(\mathbf{r}(t)) &= \langle yz, xz, xy \rangle \\ &= \langle (t^2)(t^3), (t)(t^3), (t)(t^2) \rangle \\ &= \langle t^5, t^4, t^3 \rangle \end{aligned}$$

Step 4: Compute the dot product

Now compute

$$\begin{aligned} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) &= \langle t^5, t^4, t^3 \rangle \cdot \langle 1, 2t, 3t^2 \rangle \\ &= t^5(1) + t^4(2t) + t^3(3t^2) \\ &= t^5 + 2t^5 + 3t^5 \\ &= 6t^5 \end{aligned}$$

Step 5: Integrate over the interval

Hence the work is

$$\begin{aligned} W &= \int_0^1 6t^5 dt \\ &= 6 \int_0^1 t^5 dt \\ &= 6 \left[\frac{t^6}{6} \right]_0^1 \\ &= [t^6]_0^1 \\ &= 1 - 0 \\ &= 1 \end{aligned}$$

Final Answer:

1

5.2 Double Integrals

Key Concept

A double integral over a region R is:

$$\iint_R f(x, y) dA$$

This represents accumulation over an area.

Iterated Integrals

Double integrals are computed as iterated integrals:

$$\int_a^b \int_c^d f(x, y) dx dy$$

Example

Evaluate the accumulation of the function $f(x, y) = x + y$ over the region $R = [0, 1] \times [0, 1]$.

Step 1: Conceptual understanding (before writing the integral)

We interpret this as adding up the values of $x + y$ over the entire square region.

Imagine dividing the region into many tiny rectangles of area ΔA .

The total accumulation is approximated by:

$$\sum f(x_i, y_i) \Delta A$$

As the rectangles become infinitely small, this becomes a double integral.

Step 2: Write the double integral

$$\iint_R (x + y) dA$$

Since R is a rectangle, we convert it into an iterated integral:

$$\int_0^1 \int_0^1 (x + y) dx dy$$

Step 3: Integrate with respect to x

Treat y as constant:

$$\begin{aligned} \int_0^1 (x + y) dx &= \int_0^1 x dx + \int_0^1 y dx \\ &= \left[\frac{x^2}{2} \right]_0^1 + y[x]_0^1 \\ &= \frac{1}{2} + y \end{aligned}$$

Step 4: Integrate with respect to y

$$\begin{aligned} \int_0^1 \left(\frac{1}{2} + y \right) dy &= \left[\frac{y}{2} + \frac{y^2}{2} \right]_0^1 \\ &= \frac{1}{2} + \frac{1}{2} \\ &= 1 \end{aligned}$$

Final Answer: 1

Example

We are given the function:

$$f(x, y) = 1$$

over the region:

$$R = [0, 2] \times [0, 3]$$

This region represents a rectangle in the xy -plane with:

- width in the x -direction: $2 - 0 = 2$
- height in the y -direction: $3 - 0 = 3$

Thus, geometrically, we already expect the area to be:

$$\text{Area} = 2 \times 3 = 6$$

Step 1: Interpret the double integral

The double integral

$$\iint_R 1 \, dA$$

represents the accumulation of the constant value 1 over the region R .

Conceptually, this means:

- we are summing many small pieces of area ΔA
- each piece contributes exactly its area (since $f = 1$)

So this integral directly computes the **area of the region**.

Step 2: Convert to an iterated integral

Since R is a rectangle, we write:

$$\iint_R 1 \, dA = \int_0^3 \int_0^2 1 \, dx \, dy$$

Step 3: Integrate with respect to x

Treat y as constant:

$$\int_0^2 1 \, dx = [x]_0^2 = 2$$

Step 4: Integrate with respect to y

$$\int_0^3 2 \, dy = [2y]_0^3 = 6$$

Final Answer:

6

5.3 Green's Theorem

5.3.1 Statement of Green's Theorem

Key Concept

Green's Theorem relates a line integral around a closed curve C to a double integral over region R :

$$\oint_C (P \, dx + Q \, dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

5.3.2 Interpretation

- Left side: circulation along boundary
- Right side: total rotation inside region

Example

Apply Green's Theorem to compute

$$\oint_C (y^3 \, dx + x^3 \, dy)$$

where C is the circle $x^2 + y^2 = 4$.

Step 1: Identify components

$$P = y^3, \quad Q = x^3$$

Step 2: Compute partial derivatives

$$\frac{\partial Q}{\partial x} = 3x^2, \quad \frac{\partial P}{\partial y} = 3y^2$$

Step 3: Apply Green's Theorem

$$\begin{aligned}\oint_C (Pdx + Qdy) &= \iint_R (3x^2 - 3y^2) dA \\ &= 3 \iint_R (x^2 - y^2) dA\end{aligned}$$

Step 4: Convert to polar coordinates

Since the region is a circle:

$$x = r \cos \theta, \quad y = r \sin \theta$$

Then:

$$x^2 - y^2 = r^2(\cos^2 \theta - \sin^2 \theta) = r^2 \cos(2\theta)$$

The Jacobian gives $dA = r dr d\theta$.

Step 5: Set up integral

$$3 \int_0^{2\pi} \int_0^2 r^2 \cos(2\theta) \cdot r dr d\theta = 3 \int_0^{2\pi} \int_0^2 r^3 \cos(2\theta) dr d\theta$$

Step 6: Separate integrals

$$= 3 \left(\int_0^2 r^3 dr \right) \left(\int_0^{2\pi} \cos(2\theta) d\theta \right)$$

Step 7: Evaluate

$$\int_0^2 r^3 dr = \frac{16}{4} = 4, \quad \int_0^{2\pi} \cos(2\theta) d\theta = 0$$

Final Answer:

$$\boxed{0}$$

Insight: Symmetry of the region causes cancellation.

Example

Evaluate

$$\oint_C (x^2 y dx + xy^2 dy)$$

where C bounds the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$.

Step 1: Identify components

$$P = x^2y, \quad Q = xy^2$$

Step 2: Compute derivatives

$$\frac{\partial Q}{\partial x} = y^2, \quad \frac{\partial P}{\partial y} = x^2$$

Step 3: Apply Green's Theorem

$$\oint_C (Pdx + Qdy) = \iint_R (y^2 - x^2) dA$$

Step 4: Describe the region

The triangle is bounded by:

$$0 \leq x \leq 1, \quad 0 \leq y \leq 2x$$

Step 5: Set up integral

$$\int_0^1 \int_0^{2x} (y^2 - x^2) dy dx$$

Step 6: Integrate with respect to y

$$\begin{aligned} \int_0^{2x} (y^2 - x^2) dy &= \left[\frac{y^3}{3} - x^2y \right]_0^{2x} \\ &= \frac{8x^3}{3} - 2x^3 \\ &= \frac{2x^3}{3} \end{aligned}$$

Step 7: Integrate with respect to x

$$\int_0^1 \frac{2x^3}{3} dx = \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{6}$$

Final Answer:

$$\boxed{\frac{1}{6}}$$

Example

Evaluate

$$\oint_C (e^x \cos y \, dx + e^x \sin y \, dy)$$

where C is the rectangle $0 \leq x \leq 1$, $0 \leq y \leq \pi$.

Step 1: Identify components

$$P = e^x \cos y, \quad Q = e^x \sin y$$

Step 2: Compute derivatives

$$\frac{\partial Q}{\partial x} = e^x \sin y, \quad \frac{\partial P}{\partial y} = -e^x \sin y$$

Step 3: Apply Green's Theorem

$$\iint_R (e^x \sin y - (-e^x \sin y)) \, dA = \iint_R 2e^x \sin y \, dA$$

Step 4: Set up integral

$$\int_0^1 \int_0^\pi 2e^x \sin y \, dy \, dx$$

Step 5: Separate variables

$$2 \int_0^1 e^x \, dx \int_0^\pi \sin y \, dy$$

Step 6: Evaluate

$$\int_0^1 e^x \, dx = e - 1, \quad \int_0^\pi \sin y \, dy = 2$$

Final Answer:

$$2(e - 1)(2) = \boxed{4(e - 1)}$$

6 Surface and Volume Integrals: From Flux to Global Theorems

6.1 Surface Integrals

Key Concept

The surface integral of a vector field over a surface S is:

$$\iint_S \mathbf{F} \cdot d\mathbf{S}$$

This measures the **flux** of a vector field through a surface.

A surface is parametrized as:

$$\mathbf{r}(u, v)$$

Then:

$$d\mathbf{S} = (\mathbf{r}_u \times \mathbf{r}_v) dA$$

Example

Compute the flux of $\mathbf{F} = \langle x, y, z \rangle$ through the surface $z = 4 - x^2 - y^2$, $z \geq 0$.

Step 1: Understand the surface and determine its projection

The equation of the surface is

$$z = 4 - x^2 - y^2$$

Because the surface satisfies $z \geq 0$, we impose

$$\begin{aligned} 4 - x^2 - y^2 &\geq 0 \\ x^2 + y^2 &\leq 4 \end{aligned}$$

So the projection of the surface onto the xy -plane is the disk

$$x^2 + y^2 \leq 4$$

which is a circle of radius 2 centered at the origin.

This tells us the region of integration in the plane.

Step 2: Parametrize the surface

Since the surface is already given as z in terms of x and y , the natural parametrization is

$$\mathbf{r}(x, y) = \langle x, y, 4 - x^2 - y^2 \rangle$$

Here:

- the first component is the x -coordinate
- the second component is the y -coordinate
- the third component is the corresponding z -value on the surface

So every point (x, y) in the disk $x^2 + y^2 \leq 4$ gives a point on the surface.

Step 3: Compute the partial derivatives of the parametrization

To find the oriented surface element, we compute the tangent vectors:

$$\begin{aligned}\mathbf{r}_x &= \frac{\partial}{\partial x} \langle x, y, 4 - x^2 - y^2 \rangle \\ &= \left\langle \frac{\partial x}{\partial x}, \frac{\partial y}{\partial x}, \frac{\partial}{\partial x}(4 - x^2 - y^2) \right\rangle \\ &= \langle 1, 0, -2x \rangle\end{aligned}$$

Similarly,

$$\begin{aligned}\mathbf{r}_y &= \frac{\partial}{\partial y} \langle x, y, 4 - x^2 - y^2 \rangle \\ &= \left\langle \frac{\partial x}{\partial y}, \frac{\partial y}{\partial y}, \frac{\partial}{\partial y}(4 - x^2 - y^2) \right\rangle \\ &= \langle 0, 1, -2y \rangle\end{aligned}$$

These two vectors lie tangent to the surface.

Step 4: Compute the cross product $\mathbf{r}_x \times \mathbf{r}_y$

The cross product gives a normal vector to the surface:

$$\mathbf{r}_x \times \mathbf{r}_y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -2x \\ 0 & 1 & -2y \end{vmatrix}$$

Expand the determinant:

$$\begin{aligned}
\mathbf{r}_x \times \mathbf{r}_y &= \mathbf{i} \begin{vmatrix} 0 & -2x \\ 1 & -2y \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & -2x \\ 0 & -2y \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \\
&= \mathbf{i}(0(-2y) - (-2x)(1)) - \mathbf{j}(1(-2y) - (-2x)(0)) + \mathbf{k}(1 \cdot 1 - 0 \cdot 0) \\
&= \mathbf{i}(2x) - \mathbf{j}(-2y) + \mathbf{k}(1) \\
&= \langle 2x, 2y, 1 \rangle
\end{aligned}$$

So the vector surface element is

$$d\mathbf{S} = (\mathbf{r}_x \times \mathbf{r}_y) dA = \langle 2x, 2y, 1 \rangle dA$$

This normal points upward because its k -component is positive.

Step 5: Evaluate the vector field on the surface

The field is

$$\mathbf{F}(x, y, z) = \langle x, y, z \rangle$$

But on the surface, $z = 4 - x^2 - y^2$. So substitute this into $\mathbf{F}$:

$$\mathbf{F}(\mathbf{r}(x, y)) = \langle x, y, 4 - x^2 - y^2 \rangle$$

Step 6: Compute the dot product $\mathbf{F} \cdot (\mathbf{r}_x \times \mathbf{r}_y)$

The flux integrand is:

$$\begin{aligned}
\mathbf{F} \cdot (\mathbf{r}_x \times \mathbf{r}_y) &= \langle x, y, 4 - x^2 - y^2 \rangle \cdot \langle 2x, 2y, 1 \rangle \\
&= x(2x) + y(2y) + (4 - x^2 - y^2)(1) \\
&= 2x^2 + 2y^2 + 4 - x^2 - y^2 \\
&= x^2 + y^2 + 4
\end{aligned}$$

Therefore the flux integral becomes

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_R (x^2 + y^2 + 4) dA$$

where R is the disk $x^2 + y^2 \leq 4$.

Step 7: Convert the integral to polar coordinates

Because the region R is a disk, polar coordinates are the natural choice:

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta \\x^2 + y^2 &= r^2 \\dA &= r \, dr \, d\theta\end{aligned}$$

The disk $x^2 + y^2 \leq 4$ becomes

$$0 \leq r \leq 2, \quad 0 \leq \theta \leq 2\pi$$

Substitute into the integral:

$$\iint_R (x^2 + y^2 + 4) \, dA = \int_0^{2\pi} \int_0^2 (r^2 + 4) r \, dr \, d\theta$$

Step 8: Simplify the integrand

Distribute the r :

$$(r^2 + 4)r = r^3 + 4r$$

So:

$$\int_0^{2\pi} \int_0^2 (r^3 + 4r) \, dr \, d\theta$$

Step 9: Integrate with respect to r

$$\begin{aligned}\int_0^2 (r^3 + 4r) \, dr &= \left[\frac{r^4}{4} + 2r^2 \right]_0^2 \\&= \left(\frac{2^4}{4} + 2(2^2) \right) - (0 + 0) \\&= \left(\frac{16}{4} + 8 \right) \\&= 4 + 8 \\&= 12\end{aligned}$$

Step 10: Integrate with respect to θ

$$\begin{aligned}
 \int_0^{2\pi} 12 \, d\theta &= 12[\theta]_0^{2\pi} \\
 &= 12(2\pi) \\
 &= 24\pi
 \end{aligned}$$

Final Answer:

$$\boxed{24\pi}$$

6.2 Stokes' Theorem

6.2.1 Conceptual Explanation

Key Concept

Stokes' Theorem states that the circulation of a vector field along a closed curve C is equal to the total curl (rotation) of the field over the surface S bounded by C :

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

In simple terms:

- The left side measures how much the vector field “flows around” the boundary C
- The right side measures how much the field “rotates” across the entire surface S

Key Idea:

$$\boxed{\text{Total circulation around a boundary} = \text{total curl inside the surface}}$$

This is extremely useful because:

- A complicated line integral can be replaced with an easier surface integral
- We only need the curl, not the full path traversal

Example

Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle -y, x, z \rangle$ and C is the boundary of the disk $x^2 + y^2 \leq 1$ in the plane $z = 1$.

Step 1: Understand the geometry

- The curve C is a circle of radius 1 - It lies in the plane $z = 1$ - The surface S is the

flat disk bounded by this circle

Instead of computing the line integral directly (which would be complicated), we apply Stokes' Theorem and convert it into a surface integral.

Step 2: Compute the curl of the vector field

Given:

$$\mathbf{F} = \langle -y, x, z \rangle$$

Compute:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ -y & x & z \end{vmatrix}$$

Expand determinant:

$$\nabla \times \mathbf{F} = \mathbf{i} \left(\frac{\partial z}{\partial y} - \frac{\partial x}{\partial z} \right) - \mathbf{j} \left(\frac{\partial z}{\partial x} - \frac{\partial(-y)}{\partial z} \right) + \mathbf{k} \left(\frac{\partial x}{\partial x} - \frac{\partial(-y)}{\partial y} \right)$$

Compute each term:

$$\begin{aligned} \frac{\partial z}{\partial y} &= 0, & \frac{\partial x}{\partial z} &= 0 \\ \frac{\partial z}{\partial x} &= 0, & \frac{\partial(-y)}{\partial z} &= 0 \\ \frac{\partial x}{\partial x} &= 1, & \frac{\partial(-y)}{\partial y} &= -1 \end{aligned}$$

So:

$$\begin{aligned} \nabla \times \mathbf{F} &= \langle 0, 0, 1 - (-1) \rangle \\ &= \langle 0, 0, 2 \rangle \end{aligned}$$

Step 3: Determine the surface normal

The surface is a flat disk in the plane $z = 1$.

- The upward normal vector is:

$$\mathbf{n} = \langle 0, 0, 1 \rangle$$

Thus:

$$d\mathbf{S} = \mathbf{n} dA = \langle 0, 0, 1 \rangle dA$$

Step 4: Compute the dot product

$$\begin{aligned}(\nabla \times \mathbf{F}) \cdot d\mathbf{S} &= \langle 0, 0, 2 \rangle \cdot \langle 0, 0, 1 \rangle dA \\ &= 2 dA\end{aligned}$$

So the surface integral becomes:

$$\iint_S 2 dA$$

Step 5: Evaluate the surface integral

Since the integrand is constant (2), we can factor it out:

$$\iint_S 2 dA = 2 \iint_S dA$$

But:

$$\iint_S dA = \text{Area of the disk}$$

The disk has radius 1:

$$\text{Area} = \pi(1)^2 = \pi$$

Thus:

$$\iint_S 2 dA = 2\pi$$

Final Answer:

$$\boxed{2\pi}$$

6.3 Triple Integrals

Key Concept

The triple integral of a scalar function over a volume V is defined as:

$$\iiint_V f(x, y, z) dV$$

Expanded Explanation:

A triple integral represents the accumulation of a quantity throughout a three-dimensional region.

To understand this, imagine dividing the volume V into many tiny rectangular boxes of volume ΔV . At each small box, the function $f(x, y, z)$ gives a value (such as density, temperature, or intensity).

We approximate the total accumulation by:

$$\sum f(x_i, y_i, z_i) \Delta V$$

As the size of these boxes shrinks to zero, the sum becomes the triple integral:

$$\iiint_V f(x, y, z) dV$$

Interpretations in Engineering:

- If $f(x, y, z) = 1$, the integral gives **volume**
- If $f(x, y, z)$ is density, the integral gives **mass**
- If $f(x, y, z)$ is energy density, the integral gives **total energy**

Why Change Coordinates?

For symmetric regions like spheres or cylinders, Cartesian coordinates become complicated. In such cases, we switch to:

- Cylindrical coordinates
- Spherical coordinates

because they simplify both the region and the integrand.

Example

Evaluate $\iiint_V (x^2 + y^2 + z^2) dV$ over the sphere $x^2 + y^2 + z^2 \leq 1$.

Step 1: Recognize symmetry

The region is a sphere centered at the origin:

$$x^2 + y^2 + z^2 \leq 1$$

This suggests using **spherical coordinates**, where:

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

Most importantly:

$$x^2 + y^2 + z^2 = r^2$$

So the integrand simplifies significantly.

Step 2: Determine bounds

For a unit sphere:

$$\begin{aligned}0 &\leq r \leq 1 \\0 &\leq \theta \leq \pi \\0 &\leq \phi \leq 2\pi\end{aligned}$$

Step 3: Volume element

In spherical coordinates:

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

This factor $r^2 \sin \theta$ accounts for the distortion when converting from Cartesian coordinates.

Step 4: Substitute into the integral

The integrand becomes:

$$x^2 + y^2 + z^2 = r^2$$

Thus:

$$\begin{aligned}\iiint_V (x^2 + y^2 + z^2) \, dV &= \int_0^{2\pi} \int_0^\pi \int_0^1 r^2 \cdot r^2 \sin \theta \, dr \, d\theta \, d\phi \\&= \int_0^{2\pi} \int_0^\pi \int_0^1 r^4 \sin \theta \, dr \, d\theta \, d\phi\end{aligned}$$

Step 5: Separate the integrals

Because the variables are independent:

$$= \left(\int_0^1 r^4 \, dr \right) \left(\int_0^\pi \sin \theta \, d\theta \right) \left(\int_0^{2\pi} d\phi \right)$$

Step 6: Evaluate each integral

Radial integral:

$$\int_0^1 r^4 \, dr = \left[\frac{r^5}{5} \right]_0^1 = \frac{1}{5}$$

Angular (θ) integral:

$$\int_0^\pi \sin \theta \, d\theta = [-\cos \theta]_0^\pi = (-(-1)) - (-1) = 2$$

Azimuthal (ϕ) integral:

$$\int_0^{2\pi} d\phi = 2\pi$$

Step 7: Multiply results

$$\frac{1}{5} \cdot 2 \cdot 2\pi = \frac{4\pi}{5}$$

Final Answer:

$$\boxed{\frac{4\pi}{5}}$$

6.4 Divergence Theorem

Key Concept

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \iint_S \mathbf{F} \cdot d\mathbf{S}$$

- Volume integral $\rightarrow$ total source (how much the field is “generated” inside)
- Surface integral $\rightarrow$ total flux (how much exits the boundary)

The Divergence Theorem states that the total outward flow (flux) of a vector field across a closed surface is equal to the total amount of “source” inside the volume.

Instead of computing a difficult surface integral directly, we can:

- compute the divergence (a scalar function),
- integrate it over the volume,
- and obtain the same result.

$$\boxed{\text{Flux out of surface} = \text{Total expansion inside volume}}$$

Example

Compute the flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ over the sphere $x^2 + y^2 + z^2 \leq 1$.

Step 1: Understand the problem

We are asked to compute:

$$\iint_S \mathbf{F} \cdot d\mathbf{S}$$

where S is the surface of the unit sphere.

Direct computation would require parametrizing the sphere and computing a surface integral, which is complicated.

Instead, we apply the Divergence Theorem:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

Step 2: Compute the divergence

Given:

$$\mathbf{F} = \langle x^3, y^3, z^3 \rangle$$

Compute divergence:

$$\begin{aligned}\nabla \cdot \mathbf{F} &= \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) \\ &= 3x^2 + 3y^2 + 3z^2\end{aligned}$$

Factor:

$$\nabla \cdot \mathbf{F} = 3(x^2 + y^2 + z^2)$$

Since:

$$x^2 + y^2 + z^2 = r^2$$

we get:

$$\nabla \cdot \mathbf{F} = 3r^2$$

Step 3: Convert to spherical coordinates

The region is a sphere:

$$x^2 + y^2 + z^2 \leq 1$$

So we use spherical coordinates:

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

Volume element:

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

Step 4: Substitute into the integral

$$\begin{aligned} \iiint_V (\nabla \cdot \mathbf{F}) \, dV &= \iiint_V 3r^2 \, dV \\ &= \int_0^{2\pi} \int_0^\pi \int_0^1 3r^2 \cdot r^2 \sin \theta \, dr \, d\theta \, d\phi \\ &= \int_0^{2\pi} \int_0^\pi \int_0^1 3r^4 \sin \theta \, dr \, d\theta \, d\phi \end{aligned}$$

Step 5: Separate the integrals

Because the variables are independent:

$$= 3 \left(\int_0^1 r^4 \, dr \right) \left(\int_0^\pi \sin \theta \, d\theta \right) \left(\int_0^{2\pi} d\phi \right)$$

Step 6: Evaluate each integral

Radial part:

$$\int_0^1 r^4 \, dr = \left[\frac{r^5}{5} \right]_0^1 = \frac{1}{5}$$

Angular (θ):

$$\int_0^\pi \sin \theta \, d\theta = [-\cos \theta]_0^\pi = 2$$

Azimuthal (ϕ):

$$\int_0^{2\pi} d\phi = 2\pi$$

Step 7: Multiply results

$$3 \cdot \frac{1}{5} \cdot 2 \cdot 2\pi = \frac{12\pi}{5}$$

Final Answer:

$$\boxed{\frac{12\pi}{5}}$$

Activity: Vector Calculus and Integral Theorems

Answer the following questions clearly and show complete solutions where necessary.

1. Let

$$f(x, y) = x^2 + xy + y^2.$$

Find the directional derivative of f at the point $(1, 2)$ in the direction of the vector $\langle 3, 4 \rangle$.

2. Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

where

$$\mathbf{F}(x, y) = \langle y, x \rangle$$

and

$$\mathbf{r}(t) = \langle t, t^2 \rangle, \quad 0 \leq t \leq 1.$$

3. Use Green's Theorem to evaluate

$$\oint_C (x^2 y \, dx + xy^2 \, dy)$$

where C is the positively oriented boundary of the triangle with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$.

4. Use Stokes' Theorem to evaluate

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

where

$$\mathbf{F}(x, y, z) = \langle -y, x, z \rangle$$

and C is the boundary of the disk

$$x^2 + y^2 \leq 1$$

in the plane $z = 1$.

5. Use the Divergence Theorem to compute the outward flux of

$$\mathbf{F}(x, y, z) = \langle x^3, y^3, z^3 \rangle$$

across the surface of the unit sphere

$$x^2 + y^2 + z^2 = 1.$$

Answer Key: Vector Calculus and Integral Theorems Activity

1. Let

$$f(x, y) = x^2 + xy + y^2.$$

First compute the gradient:

$$\nabla f = \langle 2x + y, x + 2y \rangle$$

At $(1, 2)$:

$$\nabla f(1, 2) = \langle 4, 5 \rangle$$

The magnitude of $\langle 3, 4 \rangle$ is

$$\sqrt{3^2 + 4^2} = 5$$

so the unit vector is

$$\mathbf{u} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$$

Hence the directional derivative is

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \langle 4, 5 \rangle \cdot \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle = \frac{12}{5} + \frac{20}{5} = \frac{32}{5}$$

$\frac{32}{5}$

2. We are given

$$\mathbf{F}(x, y) = \langle y, x \rangle, \quad \mathbf{r}(t) = \langle t, t^2 \rangle, \quad 0 \leq t \leq 1$$

Differentiate the curve:

$$\mathbf{r}'(t) = \langle 1, 2t \rangle$$

Evaluate the vector field along the curve:

$$\mathbf{F}(\mathbf{r}(t)) = \langle t^2, t \rangle$$

Compute the dot product:

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = t^2(1) + t(2t) = 3t^2$$

Therefore,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 3t^2 dt = [t^3]_0^1 = 1$$

1

3. By Green's Theorem,

$$\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

where

$$P = x^2y, \quad Q = xy^2$$

Compute the needed partial derivatives:

$$\frac{\partial Q}{\partial x} = y^2, \quad \frac{\partial P}{\partial y} = x^2$$

Hence,

$$\oint_C (x^2y dx + xy^2 dy) = \iint_R (y^2 - x^2) dA$$

The triangular region is bounded by

$$0 \leq x \leq 1, \quad 0 \leq y \leq 2x$$

so

$$\iint_R (y^2 - x^2) dA = \int_0^1 \int_0^{2x} (y^2 - x^2) dy dx$$

Evaluate the inner integral:

$$\int_0^{2x} (y^2 - x^2) dy = \left[\frac{y^3}{3} - x^2y \right]_0^{2x} = \frac{8x^3}{3} - 2x^3 = \frac{2x^3}{3}$$

Then

$$\int_0^1 \frac{2x^3}{3} dx = \frac{2}{3} \left[\frac{x^4}{4} \right]_0^1 = \frac{1}{6}$$

$\frac{1}{6}$

4. By Stokes' Theorem,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

where

$$\mathbf{F} = \langle -y, x, z \rangle$$

Compute the curl:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ -y & x & z \end{vmatrix} = \langle 0, 0, 2 \rangle$$

The surface S is the disk $x^2 + y^2 \leq 1$ in the plane $z = 1$, oriented upward, so

$$\mathbf{n} = \langle 0, 0, 1 \rangle$$

and

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = \langle 0, 0, 2 \rangle \cdot \langle 0, 0, 1 \rangle = 2$$

Therefore,

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_S 2 \, dA = 2(\text{area of unit disk}) = 2\pi$$

$$\boxed{2\pi}$$

5. By the Divergence Theorem,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

where

$$\mathbf{F} = \langle x^3, y^3, z^3 \rangle$$

Compute the divergence:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) = 3x^2 + 3y^2 + 3z^2$$

On the unit sphere, spherical coordinates are convenient:

$$x^2 + y^2 + z^2 = r^2, \quad dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

Thus,

$$\begin{aligned} \iiint_V (\nabla \cdot \mathbf{F}) \, dV &= \int_0^{2\pi} \int_0^\pi \int_0^1 3r^2(r^2 \sin \theta) \, dr \, d\theta \, d\phi \\ &= 3 \left(\int_0^1 r^4 \, dr \right) \left(\int_0^\pi \sin \theta \, d\theta \right) \left(\int_0^{2\pi} d\phi \right) \end{aligned}$$

Evaluate each integral:

$$\int_0^1 r^4 \, dr = \frac{1}{5}, \quad \int_0^\pi \sin \theta \, d\theta = 2, \quad \int_0^{2\pi} d\phi = 2\pi$$

Therefore,

$$3 \cdot \frac{1}{5} \cdot 2 \cdot 2\pi = \frac{12\pi}{5}$$

$$\boxed{\frac{12\pi}{5}}$$